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(57) Abstract :

The present invention relates to a system and method for enabling adaptive learning from presentation slides through multi-modal content generation and concept graph modelling. The system comprises a slide ingestion and preprocessing module for extracting and organizing slide content; a slide intelligence construction module for generating semantic summaries, key concepts, assessment elements, and metadata; a concept dependency graph generation module for establishing prerequisite relationships; an adaptive learning scheduling for selecting suitable learning modes based on learner performance and interaction; a multi-mode learning execution to generate different modes of learning, which include summarization, assessment, visualization, and explanation; a deterministic caching module using a content-based hashing technique for caching generated outputs; a Feedback and Adaptive Update module for updating learner profiles and knowledge gaps; and a Feedback-Driven Generative Validation Loop configured to validate generated content and refine outputs based on structural and semantic feedback. The present invention describes a framework for a personalized learning system from presentation-based educational materials, providing a clear and intelligent approach to such a system.

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